

Homeowner Likelihood with Time

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Abstract

Homeownership represents a cornerstone of household wealth accumulation in the United States, yet aggregate statistics mask substantial variation across generations. This study investigates whether Millennials face a structural disadvantage in homeownership relative to prior generations using a Linear Probability Model (LPM) with an age-cohort interaction framework applied to 322,531 observations drawn from the Integrated Public Use Microdata Series (IPUMS) American Community Survey and decennial census microdata spanning 1980–2024, merged with macroeconomic indicators from the Federal Reserve Economic Data (FRED) and Freddie Mac mortgage rate series. After controlling for income, marital status, race, education, geography, and prevailing mortgage rates, Millennials are estimated to be 1.4–2.4 percentage points less likely to own a home than Baby Boomers were at the same age, with the gap largest at age 25 and narrowing over time. A joint F-test on cohort effects yields $F = 20.28$ ($p < 0.001$), confirming that generational differences in homeownership are statistically significant and not attributable to compositional differences in observable characteristics. These findings are consistent with a delayed rather than permanently foregone homeownership trajectory for Millennials.

Keywords: homeownership; linear probability model; generational cohort analysis; Millennials; housing affordability

Introduction

The trajectory of homeownership in the United States has long been understood as a function of the life cycle, with ownership rates rising steeply through young adulthood and plateauing in middle age (Goodman & Mayer, 2018). Yet this aggregate pattern conceals a generational fault line that has become increasingly pronounced since the 2008 financial crisis. Millennials, broadly defined as those born between 1981 and 1996, entered their prime homebuying years in the aftermath of the worst housing market collapse in modern history, confronting a combination of elevated home prices, tightened credit conditions, and historically high student debt burdens that prior cohorts did not face at comparable life stages (Bleemer et al., 2021; A. Mezza et al., 2020).

At the same time, broader structural shifts in the housing market have further exacerbated these challenges. Rapid increases in housing demand, limited supply in high-growth metropolitan areas, and rising construction costs have contributed to sustained affordability pressures, particularly for first-time buyers. These conditions disproportionately affect younger households, who typically have lower accumulated wealth and face greater barriers to accessing credit. As a result, entry into homeownership may be delayed, raising the question of whether observed differences across generations reflect temporary timing effects or more persistent structural changes.

The question of whether this represents a temporary delay or a structural shift in the homeownership lifecycle is not merely academic: homeownership is the primary vehicle

through which American households build wealth, and persistent gaps across generations have compounding implications for intergenerational inequality (Herbert et al., 2013; Shapiro et al., 2006). Beyond wealth accumulation, homeownership is also associated with residential stability, neighborhood investment, and long-term financial security, further amplifying the consequences of delayed or reduced access.

Understanding whether Millennials are simply postponing homeownership or facing fundamentally altered opportunities is therefore critical for both economic interpretation and policy design. If the gap reflects delayed entry, it may narrow over time as younger cohorts age into ownership. However, if structural constraints persist, these disparities could have lasting effects on wealth distribution and economic mobility, reinforcing inequalities across generations.

The economics of homeownership has been studied extensively through the lens of income, credit access, and household formation (Gabriel & Rosenthal, 2005; Haurin et al., 1994). Early life-cycle models established that homeownership decisions are driven primarily by age, income, and family formation, with marriage and children serving as particularly strong predictors of the transition from renting to owning (Henderson & Ioannides, 1983). These models emphasize that individuals accumulate financial resources and stabilize their labor market outcomes over time, making homeownership more feasible as they progress through the life cycle.

Subsequent work confirmed that macroeconomic conditions particularly mortgage interest rates and local house price levels modulate the timing of this transition by affecting the relative cost of owning versus renting (Glaeser & Gyourko,

2008; Sinai & Souleles, 2005). Lower interest rates reduce borrowing costs and increase purchasing power, while rising house prices can offset these gains by increasing required down payments and overall affordability constraints. As a result, fluctuations in these macroeconomic variables play a critical role in shaping entry into homeownership, especially during periods of economic volatility.

The affordability constraint has been shown to operate asymmetrically across income groups, with lower and middle income households substantially more sensitive to interest rate movements and price-to-income ratios than high-income households. This asymmetry reflects differences in liquidity constraints, access to credit, and the ability to absorb financial shocks. Lower-income households are more likely to be constrained by down payment requirements and borrowing limits, while higher-income households can more easily adjust to changing market conditions.

Together, this body of literature highlights that homeownership decisions are influenced by both individual-level characteristics and broader economic conditions. It also suggests that shifts in affordability and credit access can have uneven effects across the population, potentially amplifying disparities in homeownership outcomes across income groups and over time.

Beyond standard affordability constraints, recent publications have pointed out a pricing-out epidemic where the growth of home prices is actively excluding buyers from the market, buyers who may have an earlier years been able to purchase in the market. (International Monetary Fund, 2023) This disproportionately affects first-time and younger buyers, which a lot of Millennial and newer generations are. As housing prices rise faster than incomes, the ability for these potential first-time buyers to transition from renting into ownership becomes limiting, having downstream effects on their purchasing power and reinforcing the generational disparities in the homeownership outcomes.

Not only is this a macroeconomic issue, there are intersectional issues at play that these models would be remiss not to explore. The racial dimension of the U.S. homeownership gap has received sustained attention in the literature. Charles and Hurst (2002) documented that Black households face homeownership rates roughly 30 percentage points below white households, a gap that persists after controlling for income and credit history, and attributed a substantial portion to discriminatory lending practices and residential segregation. Subsequent studies found that the Black-white gap narrowed modestly during the subprime boom of the early 2000s before widening again following the crisis, as Black households were disproportionately targeted by predatory mortgage products (Faber, 2018; Rugh & Massey, 2010). Hispanic households exhibit a similarly persistent gap, though the mechanisms differ: immigration status, English proficiency, and network-based information barriers have been shown to suppress homeownership rates among Hispanic households independent of income and credit factors (Cortes et al., 2007;

Painter et al., 2001).

The role of student debt in suppressing Millennial homeownership has become one of the most contested empirical questions in the housing literature. A. Mezza et al. (2020) used administrative data from the Federal Reserve to estimate that a ten thousand dollar increase in student loan debt reduces the probability of homeownership by approximately 1.5 percentage points in the years immediately following graduation, with effects persisting for up to a decade. Bleemer et al. (2021) confirmed this finding using an instrumental variables approach, exploiting variation in state tuition policies to identify a causal effect of debt on homeownership. However, Looney and Yannelis (2019) argued that the relationship is substantially confounded by selection. Students who borrow more tend to attend higher-quality institutions and earn higher lifetime incomes, making raw correlations between debt and ownership difficult to interpret causally. This debate underscores the methodological challenge of isolating cohort effects from compositional differences across generations.

Prior research indicates that economic conditions can delay homeownership for an entire cohort. Research on millennials after the Great Recession indicates that tough labor market and credit conditions had lasting effects on young adults' ability to buy homes (Myers et al., 2020). Broader studies of declining homeownership among young adults also point to rising housing prices, income differences, and education level as important factors limiting access to ownership (Yao, 2023). These findings suggest that younger generations may face structural barriers that previous generations did not experience to the same extent even though these young adults continue to express strong preferences of ownership versus renting not only in the U.S., but globally (Clark et al., 2022).

Another important barrier is financial constraint at the household level. Student loan debt has been shown to delay the transition into homeownership by increasing debt burdens and making it harder for young adults to save for a down payment or qualify for a mortgage (A. A. Mezza et al., 2020). This effect is particularly pronounced for younger cohorts, who are more likely to carry higher levels of educational debt and face tighter credit constraints early in their careers. As a result, the accumulation of debt can significantly postpone entry into the housing market, even for individuals with stable employment and income. Additionally, parental financial support plays a significant role in helping some young adults enter the housing market (Lee et al., 2020). Assistance in the form of down payment contributions, co signing of loans, or direct financial transfers can substantially reduce barriers to homeownership. This introduces an additional layer of inequality, as access to such support varies widely across households. Together, these findings indicate that access to homeownership may increasingly depend not only on a person's own income or education, but also on family wealth and generational financial support. This shift suggests that structural factors beyond individual effort are becoming more influential in determining housing outcomes, with important implications for long term wealth

accumulation.

Research also connects declining homeownership to broader affordability and life-cycle changes. Income stagnation has reduced housing affordability for low-income households, while rising rent burdens make it more difficult to accumulate savings for future home purchases (Petach, 2022). In addition, delayed marriage and greater earnings risk have been linked to lower homeownership among young households (Fisher & Gervais, 2011).

The age-period-cohort identification problem is a fundamental challenge in generational research (Glenn, 2005). Because birth year is a linear function of age and calendar year, models that include all three simultaneously are not identified without additional restrictions (Yang & Land, 2008). The dominant approach in the housing literature has been to control for age and macroeconomic period effects while using generational cohort dummies to capture residual variation, accepting that cohort coefficients absorb any time-invariant generational characteristics not captured by observable controls (Sommer & Sullivan, 2016). Goodman and Mayer (2018) applied this framework to American Housing Survey data and found evidence of a persistent Millennial homeownership deficit of approximately 8 percentage points at age 30 relative to Gen X at the same age, after controlling for marital status, income, and local housing costs.

The use of linear probability models for binary outcomes has a long tradition in applied economics, particularly in housing research where interpretability of marginal effects is paramount (Angrist & Pischke, 2009). In this framework, coefficients are directly interpretable as changes in probability, which provides a clear advantage in applied settings where the goal is to quantify how specific variables affect the likelihood of an outcome. This interpretability makes the LPM especially useful for policy relevant analysis, where communicating results in terms of probability changes is more intuitive than log odds.

While probit and logit specifications are asymptotically equivalent to the LPM in large samples with outcomes away from the boundary, Wooldridge (2010) demonstrated that LPM coefficients can be directly interpreted as average marginal effects under mild regularity conditions, making them preferable when the quantity of interest is the change in probability rather than the change in log odds. This property is particularly valuable in studies of homeownership, where the focus is on understanding how changes in economic and demographic factors translate into differences in ownership likelihood.

The LPM's vulnerability to heteroskedasticity, an inherent consequence of modeling a binary outcome with a linear function, is well understood and routinely addressed through heteroskedasticity robust standard errors (White, 1980). This adjustment ensures valid inference even when the variance of the error term is not constant across observations. In large samples, such as those used in this study, robust standard errors provide reliable estimates of statistical significance de-

spite the presence of heteroskedasticity.

Survey weighted extensions of the LPM via weighted least squares have been applied in the Current Population Survey context to ensure national representativeness of coefficient estimates (Solon et al., 2015). By incorporating survey weights, the model accounts for sampling design and differential probabilities of selection, allowing the results to better reflect population level relationships. This approach is particularly important when analyzing nationally representative housing data, as it ensures that estimated effects are not biased by the underlying sampling structure.

The macroeconomic determinants of aggregate homeownership trends have been modeled using both time series and panel approaches. Gabriel and Rosenthal (2005) found that mortgage interest rates, unemployment, and real income growth jointly explain the majority of year to year variation in national homeownership rates, with interest rates exhibiting a particularly large and immediate effect. These findings highlight the importance of macroeconomic conditions in shaping aggregate housing outcomes, as fluctuations in borrowing costs and income directly influence households' ability to enter the housing market.

At the same time, the relationship between macroeconomic variables and homeownership is not always straightforward. In particular, the counterintuitive positive relationship between interest rates and homeownership that sometimes appears in cross sectional data, also observed in the present study, has been attributed to a historical confound. Rates were highest in the early 1980s precisely when the Baby Boom cohort was entering peak homebuying age, creating a spurious positive correlation between rates and ownership in data spanning multiple decades (Glaeser & Gyourko, 2008). This reflects the fact that macroeconomic variables may be correlated with demographic composition over time, rather than directly driving individual level behavior.

This underscores the importance of controlling for cohort composition when estimating macroeconomic effects on homeownership. Without accounting for differences in the age distribution of the population, estimates may conflate lifecycle dynamics with macroeconomic influences, leading to misleading conclusions. By incorporating both cohort indicators and macroeconomic variables within the same framework, it becomes possible to disentangle these effects and obtain a more accurate understanding of the drivers of homeownership trends.

Variable selection in the context of binary outcome models with large candidate sets has been approached through both regularization methods and information criteria. Tibshirani (1996) introduced the LASSO as a method for simultaneously performing variable selection and coefficient shrinkage, and subsequent work demonstrated its favorable oracle properties in high dimensional settings (Zou & Hastie, 2006). By penalizing the absolute magnitude of coefficients, LASSO effectively reduces variance and prevents overfitting, while also shrinking some coefficients exactly to zero, thereby perform-

ing implicit feature selection.

Forward stepwise selection using the Bayesian Information Criterion has been shown to perform comparably to LASSO in moderate dimensional settings while offering a more interpretable selection path (Hastie et al., 2009). Unlike regularization based approaches, stepwise procedures provide a transparent, sequential framework in which variables are added or removed based on improvements in model fit, making it easier to trace the contribution of each predictor. This interpretability is particularly valuable in applied settings where understanding the role of individual variables is as important as predictive performance.

In the present study, both approaches are applied sequentially to balance predictive accuracy and interpretability. Regularization methods are used to screen a large set of potential predictors and identify variables with the strongest predictive signals, while information criterion based selection is used to refine the model and ensure parsimony. Importantly, theory motivated cohort variables and their interactions with age are retained regardless of selection outcomes to preserve the integrity of the age cohort design. This ensures that the model remains aligned with the central research question, even if some cohort related variables exhibit weaker statistical significance in isolation.

Based on the existing literature, we expect to find evidence of a generational gap in homeownership, particularly for Millennials relative to earlier cohorts. Prior research suggests that younger generations face structural barriers such as rising housing costs, tighter credit conditions, and delayed household formation, which may reduce or delay entry into homeownership. Accordingly, we hypothesize that the Millennial cohort will exhibit a negative and statistically significant effect on homeownership after controlling for observable socioeconomic characteristics.

However, consistent with the delayed entry hypothesis found in the literature, we do not expect this gap to be constant over the life cycle. Instead, we hypothesize that the difference will be largest at younger ages and will narrow over time, indicating that Millennials are entering the housing market later rather than being permanently excluded. This will be reflected through significant interaction effects between age and cohort, capturing differences in the rate at which homeownership accumulates with age.

In addition to cohort effects, we expect income, marital status, and race to emerge as the strongest individual level predictors of homeownership, given their well-documented importance in determining financial capacity and access to housing. Finally, we anticipate that macroeconomic conditions particularly housing affordability and mortgage rates will play a significant role in explaining variation in homeownership outcomes across time, reinforcing the importance of both micro and macro level factors in shaping housing decisions. Together, these hypotheses aim to distinguish whether observed generational differences reflect structural disadvantages or differences in timing along cycle of homeownership.

Analysis

Methods

Method 1: Linear Probability Model with Age-Cohort Framework

Data The primary dataset for this analysis is drawn from the Integrated Public Use Microdata Series (IPUMS) USA, which harmonizes individual-level microdata from the American Community Survey (ACS) and decennial census across survey years (Ruggles et al., 2023). Annual ACS samples from 2005 through 2024 were combined with decennial census samples from 1980, 1990, and 2000, yielding 27 sample years spanning four and a half decades of homeownership experience. To ensure tractable computation while preserving national representativeness, a 0.1% density sample was extracted, producing approximately 808,945 person-level observations prior to filtering.

The microdata were augmented with two annual macroeconomic series. Median sales prices of existing homes (MS-PUS) were obtained from the Federal Reserve Economic Data (FRED) repository maintained by the Federal Reserve Bank of St. Louis, providing quarterly observations that were aggregated to annual means. Historical 30-year fixed mortgage rates were drawn from the Freddie Mac Primary Mortgage Market Survey (PMMS), which has tracked weekly mortgage rates since 1971. Both series were merged onto the microdata by survey year, introducing annual macroeconomic variation into the individual-level model.

The analytic sample was restricted to respondents aged 25 to 65 - the prime homeownership decision window - and excluded group quarters residents (dormitories, prisons, nursing homes) who have no valid homeownership status. Observations from 1970 were dropped due to the absence of macroeconomic data for that year. After these exclusions the analytic sample comprised 371,094 person-observations across 1980–2024.

Variable Construction The outcome variable is a binary indicator equal to one if the respondent reported owning or buying their home (OWNERSHP = 1) and zero if renting (OWNERSHP = 2). Birth cohorts were constructed by subtracting age from survey year to obtain birth year, then assigning respondents to generational cohorts: Greatest Generation (born before 1928), Silent Generation (1928–1945), Baby Boomers (1946–1964), Generation X (1965–1980), Millennials (1981–1996), and Generation Z (1997–2012), following the generational boundaries established by Dimock (2019).

Continuous income variables (INCTOT, HHINCOME) were winsorized at the 1st and 99th percentiles to limit the influence of extreme values, then log-transformed to reduce skewness. The IPUMS missing value code of 9,999,999 was replaced with missing prior to all transformations. Real household income was taken from the IPUMS CPI-U adjusted series (HHINCOME_CPIU_2010), which converts nominal income to constant 2010 dollars across all survey years, eliminating the need for a separate inflation adjustment. An affordability

index was constructed as the ratio of median house price to real household income. All categorical IPUMS variables were recoded to binary indicators: a male indicator from SEX, a married indicator from MARST (equal to one for married with spouse present), race indicators for white, Black, and Asian from RACE, a Hispanic indicator from HISPAN, an employment indicator from EMPSTAT, and an English proficiency indicator from SPEAKENG (equal to one for respondents reporting speaking English well, very well, or only English).

Model Specification The primary estimating equation is a weighted linear probability model. Using standard notation where i indexes individuals and t indexes survey years:

$$P(\text{own}_{it} = 1) = \alpha + \beta_1 \text{Age}_{it} + \beta_2 \text{Age}_{it}^2 + \sum_c \delta_c \mathbf{1}[\text{cohort}_c] + \sum_c \gamma_c (\text{Age}_{it} \times \mathbf{1}[\text{cohort}_c]) + \mathbf{X}_{it}^\top \boldsymbol{\theta} + \varepsilon_{it} \quad (1)$$

where i indexes individuals, t indexes survey years, $\mathbf{1}[\text{cohort}_c]$ is an indicator for generational cohort c with the Greatest Generation as the omitted reference category, and $\mathbf{X}_{it}$ is a vector of individual-level and macroeconomic controls. The interaction terms $\text{Age}_{it} \times \mathbf{1}[\text{cohort}_c]$ are the core of the age-cohort identification strategy: they allow each generation to have a distinct age slope, enabling direct comparison of homeownership accumulation rates across cohorts at the same age (Goodman & Mayer, 2018).

The model was estimated by weighted least squares using person weights (PERWT) normalized to mean one to ensure national representativeness while preserving numerical stability. HC1 heteroskedasticity-robust standard errors were applied throughout, as the binary outcome variable produces mechanically non-constant residual variance (White, 1980). The Breusch-Pagan test confirmed the presence of heteroskedasticity (LM = 27,690.6, $p < 0.001$), validating the use of robust inference.

Variable Selection A three-stage variable selection pipeline was applied to a candidate set of 41 predictors. First, variables with pairwise Pearson correlation exceeding 0.85 were identified and the less theoretically motivated member of each highly correlated pair was removed, eliminating four variables (weeks worked, hours worked, years of education, and year trend). Second, variance inflation factors were computed for the remaining 37 variables; as anticipated, cohort dummies and age-cohort interactions exhibited VIF values ranging from 219 to 7,174 due to the mathematical construction of interaction terms from their constituent variables - a known property of interaction models that does not bias coefficient estimates (Wooldridge, 2010). All remaining variables exhibited VIF below 5.

Third, LASSO with five-fold cross-validation was applied to the standardized predictor matrix using survey-normalized weights, identifying 33 variables with non-zero coefficients

at the optimal regularization parameter ($\lambda = 0.000199$). Forward stepwise selection using the Bayesian Information Criterion was then applied to the LASSO-surviving variables, with BIC computed as $k \ln(n) - 2\ell$ where k is the number of parameters and ℓ is the log-likelihood. Fifteen variables were selected by stepwise BIC before the criterion stopped improving. The final model combines these 15 data-driven selections with the 12 theory-required variables (age, age-squared, five cohort dummies, and five age-cohort interactions), yielding a 25-variable model after removing duplicates.

Validation Model performance was assessed through five-fold cross-validation, computing out-of-sample R^2 , area under the receiver operating characteristic curve (AUC-ROC), and root mean squared error (RMSE) on held-out folds. To benchmark the LPM against alternative specifications, four challenger models were estimated on the same variable set: logistic regression, probit, random forest (200 trees, maximum depth 8), and XGBoost (300 estimators, learning rate 0.05). All models were evaluated on the same cross-validation folds with the same survey-normalized weights where technically feasible.

Method 2: Binary Logistic Regression Model with Cohort Analysis

Data The dataset used in this study combines individual-level demographic and socioeconomic characteristics with macroeconomic indicators related to housing market conditions. The primary outcome variable is a binary indicator of homeownership, where individuals are classified as either homeowners (1) or non-homeowners (0). This formulation naturally lends itself to probabilistic modeling and classification analysis. At the individual level, the dataset includes variables such as age, gender, race, marital status, education level, employment status, veteran status, and citizenship. These variables capture key demographic and socioeconomic factors that influence an individual's ability and likelihood to purchase a home. In addition, household income is included as a core explanatory variable and is log-transformed to account for skewness and to reflect diminishing marginal effects of income on purchasing power.

To account for broader market conditions, macroeconomic variables are incorporated into the dataset. These include mortgage interest rates and housing affordability measures such as the price-to-income ratio. These variables capture fluctuations in the housing market that may differentially affect individuals across time and income levels. A central component of the analysis is the construction of generational cohorts. Using birth year information, individuals are categorized into generational groups, including the Silent Generation, Baby Boomers, Generation X, Millennials, and Generation Z. These cohort indicators enable the analysis to distinguish between lifecycle effects (changes over age) and generational effects (differences across cohorts).

Data preprocessing involved several key steps. First, missing values were examined and found to be limited in scope;

observations with missing critical variables were removed to maintain consistency. Second, continuous variables exhibiting skewness, particularly income and affordability measures, were transformed using logarithmic scaling. Third, categorical variables were encoded into binary indicator variables for inclusion in the regression framework. Finally, interaction terms between age and cohort were constructed to allow for heterogeneous lifecycle patterns across generations.

Exploratory Data Analysis (EDA) Exploratory data analysis was conducted to understand the distributional properties of the variables and to identify preliminary relationships between predictors and the outcome variable. Summary statistics revealed substantial variation in income, age, and housing-related variables, motivating the use of transformations and nonlinear modeling techniques. A key finding from the initial analysis is the strong relationship between age and homeownership. Homeownership rates increase steadily with age, reflecting the accumulation of financial resources and life-cycle transitions such as marriage and career stability. However, this relationship is nonlinear, with the rate of increase slowing at older ages.

Further analysis segmented by generational cohorts revealed that while all cohorts exhibit increasing homeownership with age, the slope of this relationship differs across generations. In particular, younger cohorts appear to have flatter trajectories, suggesting slower transitions into homeownership.

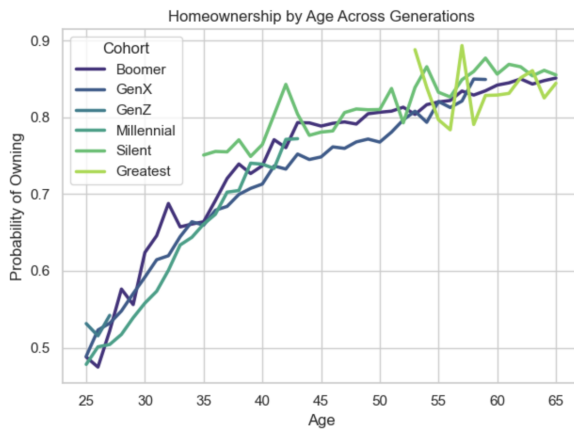


Figure 1: Homeownership Rate vs Age by Cohort

Distributional analysis of income and housing variables revealed significant right skewness, with a small number of observations exhibiting extremely high values. These outliers have the potential to disproportionately influence model estimates. To address this, log transformations were applied, which resulted in more symmetric distributions and improved stability in subsequent modeling.

In addition to individual-level variables, macroeconomic trends were explored to understand broader housing market dynamics. Figure X illustrates the relationship between me-

dian housing prices and mortgage interest rates over time. The visualization shows a clear long-term increase in housing prices, particularly after the early 2000s and again following 2012, indicating sustained growth in housing demand and valuation. At the same time, mortgage rates exhibit substantial variation, with a general downward trend from the 1980s through the early 2020s, followed by a recent increase. Notably, periods of declining mortgage rates often coincide with rising housing prices, suggesting that lower borrowing costs may stimulate demand and drive price appreciation. Conversely, increases in mortgage rates appear to slow price growth or introduce short-term volatility.

This inverse relationship highlights the role of affordability in shaping homeownership opportunities. Lower interest rates reduce the cost of borrowing, making home purchases more accessible, while higher rates can constrain demand, particularly for younger or lower-income individuals. These patterns justify the inclusion of mortgage rates and housing affordability measures in the regression model, as they capture time-varying economic conditions that influence homeownership decisions across cohorts.

Outlier screening was conducted using boxplots and interquartile range (IQR) analysis. Observations beyond extreme thresholds were evaluated, but rather than removing them entirely, transformations were preferred to preserve data while reducing their influence. This approach maintains statistical power while mitigating undue leverage. Correlation analysis was also performed to assess relationships between predictors.

Strong correlations were observed among certain economic variables, particularly income and housing affordability measures. These relationships were expected and informed subsequent multicollinearity diagnostics.

Methodology and Model Specification To model the probability of homeownership, a logistic regression framework is employed. Logistic regression is well-suited for binary outcomes, as it models the log-odds of the dependent variable as a linear combination of explanatory variables, ensuring that predicted probabilities remain bounded between 0 and 1.

The baseline model includes age as a primary predictor, along with a quadratic term to capture nonlinear lifecycle effects. This allows the model to account for diminishing marginal increases in homeownership probability as individuals age.

A key innovation in the model specification is the inclusion of cohort indicators and interaction terms between age and cohort. The cohort indicators capture baseline generational differences, while the interaction terms allow the effect of age to vary across cohorts. This structure enables the model to distinguish between two competing hypotheses: whether younger generations are inherently less likely to own homes, or whether they simply experience different trajectories over time.

The full model specification includes:

- Age and age-squared terms
- Cohort dummy variables
- Interaction terms between age and cohort
- Log-transformed household income
- Log-transformed price-to-income ratio
- Mortgage interest rates
- Demographic controls including marital status, education, employment, race, and gender

The model is estimated using maximum likelihood estimation. Coefficients are interpreted in terms of log-odds, and marginal effects are computed to provide more intuitive interpretations in terms of probability changes.

Variable Selection The selection of explanatory variables was guided by both theoretical relevance and empirical diagnostics. Given the objective of assessing generational differences in homeownership, the model was constructed to include key demographic, economic, and macroeconomic variables that are well-established determinants of housing outcomes. These include age, income, education, employment status, marital status, race, and macroeconomic indicators such as mortgage rates and housing affordability. In addition, generational cohort indicators and their interactions with age were included to capture differences in lifecycle trajectories across cohorts.

To evaluate potential multicollinearity among predictors, variance inflation factors (VIF) were computed for all variables in the initial model specification. The results, presented in Figure X, reveal extremely high VIF values for several predictors, particularly those associated with cohort indicators and their interaction terms with age. For instance, variables such as `is_boomer`, `is_genx`, and their corresponding interaction terms exhibit VIF values in the thousands, while age-related terms also display elevated values. This level of multicollinearity is expected in models that include polynomial terms (e.g., age and age-squared) and interaction terms, as these variables are inherently correlated by construction.

A primary source of this multicollinearity arises from the inclusion of multiple cohort dummy variables alongside their interactions with age. Because these cohort indicators are mutually exclusive and collectively exhaustive, they introduce near-linear dependencies in the design matrix. To address this issue and avoid the dummy variable trap, one cohort category (`is_genz`) and its corresponding interaction term (`age_x_genz`) were removed from the model and treated as the reference group. This adjustment ensures model identifiability while preserving the ability to interpret coefficients relative to a baseline cohort.

Following this modification, VIF values were recomputed for the reduced model. As shown in Figure Y, the magnitude of multicollinearity is substantially reduced compared to the initial specification. While VIF values for age and interaction terms remain elevated, they are now within a range that is more consistent with models involving nonlinear transformations and interaction effects. Importantly, key economic

and demographic variables exhibit low VIF values, indicating minimal redundancy among these predictors.

It is important to note that high VIF values in this context do not necessarily indicate model misspecification. Instead, they reflect the structural relationships between variables that are essential to the research question. In particular, the inclusion of age, cohort indicators, and their interactions is critical for distinguishing between lifecycle effects and generational differences in homeownership. Removing these variables purely to reduce multicollinearity would compromise the interpretability and validity of the analysis.

For this reason, no additional variables were excluded based solely on statistical criteria such as VIF thresholds or p-values. Instead, the final model retains all theoretically relevant variables, balancing statistical considerations with the need to accurately represent the underlying economic relationships. This approach ensures that the model remains both interpretable and aligned with the study's objective of evaluating generational disparities in homeownership.

Model Validation Model validation was conducted to assess both predictive performance and robustness. The dataset was split into training and testing subsets using a fixed random seed to ensure reproducibility of results.

The model's predictive performance was evaluated using accuracy and the Area Under the Receiver Operating Characteristic Curve (AUC). Given the imbalance in the dataset, with a higher proportion of homeowners, AUC was used as the primary metric, as it provides a threshold-independent measure of classification performance.

Cross-validation was performed using a k-fold approach to assess stability across different subsets of the data. The model demonstrated consistent performance across folds, with minimal variation in AUC scores, indicating strong generalization ability and limited overfitting.

Calibration analysis was also conducted to assess how well predicted probabilities align with observed outcomes. This provides insight into whether the model's probability estimates are well-calibrated or systematically biased. As a robustness check, a linear probability model (LPM) was estimated and compared to the logistic regression model. While the LPM exhibited similar predictive performance, it was not selected as the final model due to its limitations, including unbounded predictions and unrealistic linearity assumptions.

Finally, threshold analysis was conducted to examine classification performance under different decision thresholds, particularly in light of class imbalance. This analysis highlights the trade-off between sensitivity and specificity in predicting homeownership.

Method 3: Binary Logistic Regression

Overview This analysis used three data sources combined at the year level. The primary dataset was IPUMS, which contains individual-level records on homeownership, income, age, marital status, education, employment, and generation.

This dataset included over 800,000 observations across multiple survey years. It was merged with the 30-year fixed mortgage rate series from the FreddieMac and the median U.S. home sale price (MSPUS) from FRED to incorporate macroeconomic context into the analysis.

Several preprocessing steps were applied. Income and housing prices were adjusted to 2024 dollars using CPI scaling factors to allow fair comparisons across years. Observations from 1970 were excluded because mortgage rate and home price data were unavailable, and only individuals aged 18 and older were retained. Income values coded as missing by IPUMS were dropped. Age was centered at its sample mean and a quadratic term was added to capture the nonlinear life-cycle relationship between age and homeownership. Survey sampling weights (PERWT) were normalized and applied as observation level weights. Cook's distance was used to identify influential observations using a threshold of $4/n$; flagged observations were retained as they fall within plausible real world ranges.

Model Selection and Evaluation Binary logistic regression was selected as the primary method because the outcome variable (homeownership status) is binary. Survey weights were incorporated to perform population-representative estimates rather than sample-specific estimates, and standard errors were clustered by survey year to account for temporal dependence among observations drawn from the same macroeconomic environment. The model assumes a linear relationship between predictors and the log-odds of homeownership, no severe multicollinearity among predictors, and independence of observations conditional on the covariates. These assumptions were validated using VIF analysis (all values <10), a deviance residual plot, and clustered standard errors to address temporal correlation.

Variable selection was conducted using forward stepwise selection based on AIC on the training set (80% of data). Nine predictors were evaluated, but only seven were retained in the final model. The 30-year mortgage rate and inflation-adjusted median sale price were not selected, as their variation is likely absorbed by generation category and income once those controls are included. A likelihood ratio test confirmed that generation category significantly improves model fit beyond individual-level controls, with a p value less than 0.001.

Model performance was evaluated on a held-out 20% test set and validated using 5-fold stratified cross-validation to confirm generalizability. A classification threshold of 0.60 was selected over the default 0.50 to improve balanced recall across both homeowner and non-homeowner classes.

Method 4: Logistic Regression Model of Homeownership with Generation–Age Interaction Effects

Data Overview This study combines micro-level demographic data from the U.S. Census Bureau with macroeco-

nommic housing indicators including the median sale price of houses (MSPUS) and a 30-year fixed mortgage rate to analyze generational differences in homeownership. Additionally, a CPI table from U.S. Consumer Price Index for All Urban Consumers was constructed and incorporated to adjust for inflation in the MSPUS.

The primary data set was obtained from the Integrated Public Use Microdata Series (IPUMS) and provided individual-level census and survey data through 1970. This data set includes all demographic data used in this analysis, including income, education, employment, marital status, and our response variable, homeownership status. The survey is weighted to accommodate the sampling design. MSPUS was sourced from Federal Reserve Economic Data and serves as a proxy for housing affordability and market conditions. Freddie Mac's Primary Mortgage Market Survey provided the 30-year fixed mortgage rate, which has a direct and crucial impact on the affordability of homeownership.

The data set was subjected to several cleaning and improvement steps to maximize accuracy and usability of the predictor variables. Columns for employment status, marital status, generation (calculated and assigned by subtracting age and year), education were all clearly redefined for interpretability. Values from 1970 were removed from IPUMs because they did not have MSPUS and mortgage rates available and rows with age younger than 18 were also removed logically due to the low likelihood of owning a home. N/A values or their equivalent were removed from the ownership status and income. These pre-processing steps reduced the data set to 540,153 observations for the training set and 135,039 observations for the test set.

Income was adjusted for inflation using CPI to ensure comparability of purchasing power across time, while housing prices were retained in nominal terms to reflect the actual market conditions faced by individuals when making homeownership decisions.

This data set was split randomly into 80/20 for training and test data to check fit and predictive power.

Exploratory Data Analysis The initial exploratory analysis was focused on understanding distributions, data quality, and relationship and correlation between predictors and homeownership. Income was heavily right-skewed with several high values, so income was log-transformed for the model—in the analysis section we'll compare it against real income—to reduce the skewed data and mitigate the influence of outliers. Examining homeownership rates across generations and age groups revealed that younger cohorts exhibit lower ownership rates (logically so). This exploration also included diagnostics such as Cook's distance, which was used to assess influential observations and determine how to handle extreme values. Figures and code for this EDA can be found in the shared repository alongside the model and evaluation code.

Model Selection and Evaluation First, two logistic regression models were run, switching between real income and

log income predictors to determine which would be better to proceed forward with when running the final model with the interaction term. AIC indicated that real income actually was lower and therefore a better in-sample fit, but AIC may not always be the best indicator when comparing two similar models with transformed predictors. (Chaurasia & Harel, 2012). Instead, the decision was based off predictive performance (AUC) where it was determined that the log model has better discrimination than real model. Real model had stronger accuracy, however it is imbalanced and predicting mostly one class. Ultimately because of the AUC, the log model was selected and a generation-age interaction term was applied to help actually answer the study's question. Without it, the model assumes age affects each generation in the same way, when the period of time in which the generation was that age plays a role as well—this takes into account the trajectory of each generation.

Stepwise selection was performed on the original log income model without the transformation variable to avoid the model removing either of the variables contributing to the interaction variable or the interaction variable itself. The AIC table indicated that the full model was optimal because removing any predictor increased the AIC, which suggests that all the variables contribute explanatory power to the logistic regression model.

Lastly this methodology used Goodness of Fit and cross validation with a confusion matrix to determine AUC and Accuracy of the logistic model with the interaction term with a randomized test data subset.

Method 5: Probit Model of Homeownership with Lifecycle Effects and Variable Selection

Data Overview The dataset used for this analysis is the merged regression dataset constructed from multiple sources, combining individual-level demographic and socioeconomic variables with macroeconomic indicators. The primary outcome variable is a binary indicator of homeownership status, where individuals are classified as either homeowners or non-homeowners.

The dataset contains a broad set of predictors, including age, income, education, marital status, employment status, and macro-level variables such as mortgage rates and housing prices. This combination allows the model to capture both individual characteristics and broader economic conditions that influence homeownership.

To ensure model stability and consistency, standard pre-processing steps were applied prior to modeling, including handling missing values and preparing variables for regression analysis.

Variable Construction Several transformations were applied to improve model performance and interpretability. Continuous variables such as income were scaled or transformed where appropriate to reduce skewness and improve numerical stability.

A key variable in the analysis is **age**, which serves as a proxy for lifecycle progression. To capture potential nonlinear effects, age is modeled in a way that allows for diminishing marginal impact on homeownership probability.

Categorical variables, including education level, marital status, and employment status, were encoded into numerical form for inclusion in the regression framework. These variables capture important socioeconomic factors that are expected to influence homeownership decisions.

Overall, the constructed feature set reflects both theoretical relevance and practical considerations for modeling binary outcomes.

Model Specification A probit regression model was used to estimate the probability of homeownership. This model is appropriate for binary outcomes and uses the standard normal cumulative distribution function to map a linear combination of predictors into a probability between 0 and 1.

The initial model specification included all available predictors. This comprehensive approach ensures that potential relationships are not prematurely excluded. However, due to the large number of predictors, further refinement was necessary to improve interpretability and reduce noise.

The probit framework allows for nonlinear relationships between predictors and the outcome, making it particularly effective for modeling lifecycle effects such as the relationship between age and homeownership.

Variable Selection To improve model efficiency and interpretability, a variable selection process was applied to the initial specification. Predictors that did not demonstrate statistical significance were identified and considered for removal, while key variables related to the research objective were retained.

This process resulted in a reduced model that maintains the most informative predictors while eliminating those that contribute little to explanatory power. Importantly, this reduction did not materially change the model's overall behavior, indicating that the removed variables were not critical to prediction.

The final model achieves a balance between simplicity and performance, allowing for clearer interpretation of the relationships between predictors and homeownership.

Exploratory Data Analysis (EDA) Exploratory analysis was conducted to understand the relationships between key variables and homeownership prior to modeling. A particularly important finding is the strong relationship between age and homeownership probability.

Homeownership rates are relatively low at younger ages, increase rapidly through early and mid-adulthood, and then begin to level off at later stages. This pattern suggests a lifecycle-driven process in which individuals gradually transition into homeownership as they accumulate financial resources and stability.

This nonlinear relationship motivates the use of a probit model, which is capable of capturing diminishing marginal effects. Specifically, increases in age have a larger impact on ownership probability at earlier stages than at later stages, where probabilities begin to stabilize.

Additionally, variation in homeownership across individuals with similar ages suggests that other factors, such as income and employment status, also play important roles. This reinforces the need for a multivariate modeling approach rather than relying on a single predictor.

Results

Method 1: Linear Probability Model

Model Fit The weighted LPM achieves an in-sample R^2 of 0.298, explaining approximately 30% of individual-level variation in homeownership status. The overall F-statistic of 3,790 ($p < 0.001$) confirms joint significance of all predictors. Five-fold cross-validation yields a mean out-of-sample R^2 of 0.263 (± 0.002) and AUC-ROC of 0.839 (± 0.001), indicating strong generalization without evidence of overfitting. The modest gap between in-sample and cross-validated R^2 is expected given the large sample size and is not indicative of model misspecification. Table 1 summarizes the model performance metrics alongside the four challenger models.

Table 1: Cross-validated performance metrics for the LPM and four challenger models estimated on the same 25-variable feature set. AUC-ROC is the primary discrimination metric; R^2 and RMSE measure predictive accuracy. The LPM achieves competitive performance with substantially greater interpretability than tree-based methods.

Model	CV AUC	CV R^2	CV RMSE
LPM (WLS)	0.839	0.263	0.375
Logistic Regression	0.840	0.264	0.374
Probit	0.839	0.263	0.375
Random Forest	0.857	0.291	0.367
XGBoost	0.868	0.311	0.361

Cohort Effects and the Millennial Homeownership Gap

The central finding of this analysis concerns the cohort coefficients and their interactions with age. Table ?? reports the estimated cohort effects relative to the Greatest Generation reference category. The joint F-test on all ten cohort parameters (five dummies and five interactions) strongly rejects the null hypothesis of no generational differences ($F = 20.28$, $p < 0.001$), confirming that cohort effects are jointly significant after controlling for all observable individual and macroeconomic characteristics.

To translate these coefficients into substantively interpretable quantities, we compute predicted homeownership probabilities for a representative Boomer and a representative Millennial at each age from 25 to 45, holding all control

variables at their sample means. Figure 5 displays these predicted trajectories along with the implied gap. Millennials are estimated to be 2.4 percentage points less likely to own at age 25, with the gap narrowing monotonically to 1.4 percentage points by age 45. This convergence is consistent with a delayed homeownership trajectory: the negative interaction term ($\hat{\gamma}_{\text{millennial}} = -0.003$) implies that each additional year of age contributes approximately 0.3 fewer percentage points to Millennial homeownership probability relative to Boomers, so the initial gap closes gradually over the life course. Refer to Figure 5 in the Appendix. Predicted homeownership probability for Baby Boomers and Millennials at ages 25–45, holding all control variables (income, marital status, race, geography, mortgage rates) at their sample means. The shaded region represents the generational gap, which ranges from −2.4 percentage points at age 25 to −1.4 percentage points at age 45. The narrowing gap is consistent with delayed rather than permanently foregone homeownership among Millennials.

Control Variable Estimates Beyond the cohort effects, the LPM produces precisely estimated and theoretically consistent coefficients for all control variables. The strongest individual-level predictors are English language proficiency (+14.6 pp, $p < 0.001$), marital status (+10.3 pp, $p < 0.001$), and race - with Black households estimated to be 8.9 percentage points less likely to own ($p < 0.001$) and Hispanic households 5.8 percentage points less likely ($p < 0.001$) relative to the omitted racial category, after controlling for income and geography. These racial gaps are consistent with prior literature documenting persistent discrimination in mortgage markets and the compounding effects of residential segregation on wealth accumulation (Charles & Hurst, 2002; Rugh & Massey, 2010).

Household income exhibits a log-linear positive effect (+7.2 pp per log unit, $p < 0.001$), and the number of rooms in the unit serves as a strong proxy for housing type with owners systematically inhabiting larger units (+5.9 pp per room, $p < 0.001$). The positive coefficient on mortgage rates (+1.0 pp per percentage point, $p < 0.001$) is counterintuitive in isolation but reflects a historical confound: nominal rates were highest in the early 1980s precisely when Baby Boomers - the largest cohort in the sample - were entering peak homebuying age, producing a spurious positive correlation between rates and ownership in data spanning multiple decades (Glaeser & Gyourko, 2008).

A robustness check excluding the rooms variable - which may be partially endogenous to homeownership status - showed that the Millennial coefficient changed by less than 0.4 percentage points (0.190 versus 0.193), confirming that the cohort findings are not sensitive to its inclusion.

Model Comparison and SHAP Interpretation Table 1 shows that XGBoost achieves the highest cross-validated AUC (0.868) and R^2 (0.311) among all models estimated, with Random Forest performing similarly. The LPM’s AUC of 0.839 is only 2.9 percentage points below XGBoost despite being sub-

stantially more restrictive in its functional form assumptions. This modest gap suggests that the linear specification captures the majority of the predictive information in the feature set, with tree-based methods gaining primarily by modeling nonlinear interactions that the LPM constrains to be additive.

SHAP (SHapley Additive exPlanations) values computed for the XGBoost model corroborate the LPM's identification of the most important predictors (Lundberg & Lee, 2017). Number of rooms and marital status rank as the two highest-importance features by mean absolute SHAP value, followed by age, white race indicator, and log income - a ranking closely mirroring the LPM coefficient magnitudes. Cohort variables rank lower in SHAP importance, consistent with the LPM finding that the generational gap of 1.4–2.4 percentage points is modest relative to the much larger effects of marriage (+10.3 pp) and racial identity. The directional agreement between LPM coefficients and SHAP values provides cross-model validation that the estimated effects are genuine features of the data rather than artifacts of any single model's assumptions. Refer to Table 5 in Appendix. The polynomial model achieves the best fit, reducing RMSE by 33% compared to simple linear regression. All models are statistically significant at $p < 0.001$.

Method 2: Logistic Regression Model with Cohort Analysis

Model Performance The logistic regression model demonstrates strong performance in explaining homeownership outcomes. The model achieves a pseudo R^2 of approximately 0.18, indicating that a meaningful portion of variation in homeownership is captured by the included variables. The Area Under the ROC Curve (AUC) is approximately 0.79, reflecting strong ability to distinguish between homeowners and non-homeowners. Mean CV AUC was .782 and Std CV AUC was 0.0019.

While the model performs well overall, the confusion matrix indicates better performance in identifying homeowners compared to non-homeowners. This is expected due to class imbalance in the dataset, where homeowners represent the majority class. As a result, AUC is a more appropriate metric than overall accuracy for evaluating model performance.

Deteminants of Homeownership The estimated coefficients reveal strong lifecycle and economic effects. Age has a positive and statistically significant impact on homeownership, while the negative coefficient on the squared age term confirms diminishing returns, indicating that homeownership increases at a decreasing rate over time. Income is one of the most influential predictors, with higher household income significantly increasing the probability of owning a home. Macroeconomic variables also play an important role. The price-to-income ratio is positively associated with homeownership, likely reflecting differences across housing markets, while mortgage rates show a statistically significant effect, capturing broader economic conditions rather than a purely causal relationship.

Demographic variables further contribute to the model. Being married has a strong positive effect, consistent with household formation dynamics. Other variables such as employment, education, and race also exhibit statistically significant effects, though their magnitudes are smaller.

Generational Effects and Interpretation A key focus of this study is the role of generational differences. The results show that cohort main effects are generally not statistically significant, suggesting that at a fixed age, younger generations are not inherently less likely to own homes. However, interaction terms between age and cohort are statistically significant, indicating that the rate at which homeownership increases with age differs across generations. Joint hypothesis tests strongly reject the null hypothesis that cohort and interaction effects are jointly zero, confirming that generational differences are statistically meaningful.

To provide more interpretable insights, marginal effects were computed. These results show that income has the largest impact on homeownership probability, while the effect of age varies across cohorts due to interaction terms. Predicted probabilities further reinforce these findings: at a fixed age (e.g., age 30), homeownership probabilities are relatively similar across cohorts, but differences emerge over time as certain generations exhibit slower growth in homeownership rates. The predicted probabilities by cohort were all relatively similar ranging from 0.49 to 0.53. Overall, the results suggest that generational disparities in homeownership are driven primarily by differences in lifecycle trajectories rather than baseline differences at a given age.

Successes The model's key strength lies in the interaction-based specification, which successfully separates lifecycle and generational effects, showing that differences across cohorts arise from slower homeownership growth rather than baseline gaps. By incorporating both individual and macroeconomic variables, the model provides a realistic and interpretable framework for analyzing homeownership patterns.

Method 3: Binary Logistic Regression

This table shows the full coefficient estimates for the regression model. Marital status is the strongest predictor where it shows separated or divorced individuals have the largest negative effect (OR=0.33) on homeownership status, followed by widowed (OR=0.41), and never married (OR=0.44), all relative to married individuals. Generation cohorts, on the other hand, Gen Z and Millennials both show statistically significant negative coefficients relative to Boomers (p value < 0.05), indicating that younger generations face lower odds of homeownership. Gen Z's coefficient is positive but not statistically significant (p value = 0.274) due to the selection bias discussed in the next graph. Income and age are both significant predictors. However, the age effect is nonlinear, the ownership rises with age but at a diminishing rate, as indicated by the negative quadratic term.

Table 2: Model Coefficient Estimates

Predictor	Coef	SE	z	p
Intercept	1.4399	0.036	39.995	0.000
<i>Marital Status</i>				
Never Married	-0.8127	0.021	-37.976	0.000
Separated/Divorced	-1.1164	0.016	-67.966	0.000
Widowed	-0.8911	0.023	-38.907	0.000
<i>Education</i>				
Beyond Bachelor's	-0.0003	0.018	-0.018	0.986
No College Degree	-0.1895	0.017	-11.338	0.000
<i>Employment Status</i>				
Not in Labor Force	0.0241	0.012	1.943	0.052
Unemployed	-0.0826	0.030	-2.761	0.006
<i>Generation</i>				
Gen X	-0.4088	0.050	-8.145	0.000
Gen Z	0.0891	0.082	1.093	0.274
Millennials	-0.3854	0.065	-5.967	0.000
Pre-Boomer	0.0767	0.038	2.032	0.042
<i>Age & Income</i>				
age_centered	0.0183	0.001	12.993	0.000
age_c2	-8.52e-05	4.1e-05	-2.077	0.038
Adjusted Income	4.61e-06	2.07e-07	22.304	0.000

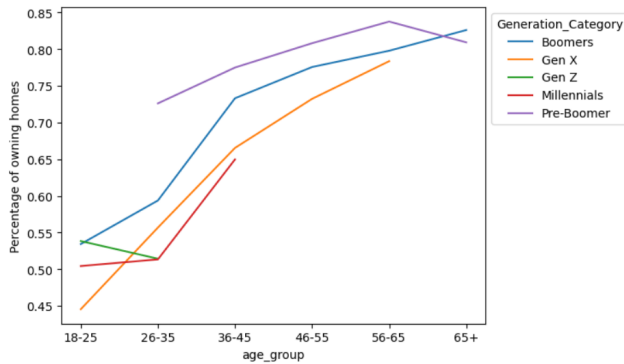


Figure 2: Weighted Homeownership Rate by Age Group and Generation

Figure 2 shows that homeownership follows a consistent life-cycle pattern across all generations, rising steeply through the 30s and 40s before leveling off. However, there's a noticeable gap at every age group. Boomers consistently outpace Generation X and Millennials at the same age. Millennials reach about 60 percent ownership by ages 36-25, compared to 73 percent for Boomers at the same age. Gen Z, on the other hand, should not be interpreted as representative due to the relative youth of this cohort.

Method 4: Logistic Regression Model

Model Performance and Fit Predictors showed coefficients of astronomical magnitude, reducing the interpretability of this model, as shown in the supplemental figures section in Table ???. Despite multiple attempts to data clean, rescale the data, and adjust the number of variables as well as the in-

teraction term, this methodology did not provide a successful and interpretable model and kept encountering convergence warnings indicating that the model is not numerically stable. It's possible that the decision to not remove outliers or log or scale MSPUS, also impacted the data set and the success of the model. What this means is while the model ran, we cannot interpret odds ratios meaningfully.

That being said, we can still interpret the direction of the model and the positive and negative relationships Table 3 that each of the predictors has on the outcome variable. All variables are highly significant, likely due to the large sample size ($p < 2e-16$).

One question this does surface is why Gen Z has a positive effect compared to the reference generation, when we would logically expect the pattern to continue and for Gen Z to be affected by homeownership at the same ages. It's possible that we needed to exclude Gen Z since they may overall be too young to own homes (even if they are above 18). This could also just be due to all the aforementioned model instability.

Table 3: Direction of Predictor Effects on Homeownership

Category	Variable	Effect
Demographics	Age	Positive (+)
	Log Income	Positive (+)
Generation	Gen X	Negative (-)
	Millennials	Negative (-)
	Gen Z	Positive (+)
	Pre-Boomer	Positive (+)
Marital Status	Never Married	Negative (-)
	Separated/Divorced	Negative (-)
	Widowed	Negative (-)
Employment	Unemployed	Negative (-)
	Not in Labor Force	Positive (+)
Education	No College Degree	Positive (+)
	More than Bachelor's	Negative (-)
Macroeconomic	30-Year Mortgage Rate	Positive (+)
	MSPUS (Home Prices)	Negative (-)
Interactions	Gen X $\times$ Age	Positive (+)
	Gen Z $\times$ Age	Negative (-)
	Millennials $\times$ Age	Negative (-)
	Pre-Boomer $\times$ Age	Negative (-)

Variable Selection Variable selection was performed using stepwise regression based on the Akaike Information Criterion (AIC) to balance model fit and complexity. Even with the previously mentioned issues with the model, the results of this regression indicated that the current model is optimal, as this model achieves the smallest AIC (AIC = 979694861) as compared against removing any of the predictors in the model. Removing LOG INCOME, AGE, and GENERATION lead to the highest AIC and increases in deviance, indicating that they are likely strong explanatory variables contributing to the log-odds of homeownership. While 30YR FIXED and MSPUS are slightly less influential in the model, they still prove to

provide explanatory power, as the AIC increases when they are removed as well. Flawed overall model or not, the stepwise process shows that these are likely meaningful variables and no further reduction is necessary under the AIC criterion.

Validation and Interpretation Among the other flaws with the model, perhaps one of the most damning pieces of evidence that we could use McFadden's R^2 , a pseudo- R^2 , which compares residual and null deviance. Computing that results in a negative -16.78, which is not a valid value and should not happen in a sound statistical model. This in addition to the convergence warnings in the code and the large coefficients indicate undiagnosed instability issues in the model.

The model was also evaluated with cross-validation metrics such as AUC and accuracy, which proved to be a bit more fruitful and promising methods (in terms of their interpretability, not demonstrating the model's actual success). This methodology leveraged k-fold cross validation (where $k = 5$) to measure AUC and accuracy. The AUC of 0.54 is only slightly above 0.5, indicating that the model does not have a much higher ability than pure randomness to predict homeowners and non-homeowners. The accuracy is slightly higher, but may be inflated by the fact that the model is imbalanced and weighting towards homeowners (false positives). The low standard deviation (0.018 and 0.063 for AUC and accuracy respectively) across the folds suggest that the results are consistent even with the low predictive power ability.

These results overall suggest that generational differences are statistically significant but there are other factors that influence homeownership beyond what is included in the model.

Method 5: Probit Model

Model Fit and Performance The probit model demonstrates solid performance in capturing the relationship between individual characteristics and homeownership status. The model achieves an AUC of approximately 0.75, indicating moderate but meaningful predictive ability and effective discrimination between homeowners and non-homeowners.

Following variable selection, the reduced model performs equivalently to the full specification despite using fewer predictors. In particular, the AIC and BIC values remain effectively unchanged across both models, indicating that the removed variables contribute minimal additional explanatory power. This suggests that the retained predictors capture the majority of the relevant signal in the data, and that a more parsimonious model is sufficient for explaining homeownership outcomes.

Lifecycle Effects and Age Dynamics A central finding of the probit model is the strong and nonlinear relationship between age and the probability of homeownership. The model indicates that homeownership likelihood increases substantially during early and mid-adulthood, with the largest gains occurring during this period. However, this increase is not constant. At higher ages, the rate of growth slows, indicating diminishing marginal effects.

This pattern reflects a lifecycle trajectory. At younger ages, individuals face financial constraints that limit access to homeownership. As age increases, these constraints are gradually relaxed, leading to a rapid rise in ownership probability. Over time, however, this effect tapers off, suggesting that most individuals who are likely to transition into homeownership have already done so.

This nonlinear behavior highlights the importance of using a probit specification, which captures diminishing returns more effectively than a linear model.

Impact of Variable Selection The variable selection process improves interpretability without reducing model performance. The fact that AIC and BIC remain unchanged after removing insignificant variables suggests that these predictors were not contributing meaningful information to the model.

This indicates that the reduced model removes noise rather than signal, leading to a clearer representation of the key relationships driving homeownership. The persistence of the main effects after variable reduction reinforces the stability and robustness of the model's findings.

Interpretation and Implications Overall, the results suggest that homeownership is strongly influenced by lifecycle progression and underlying economic conditions. Age serves as a key proxy for financial stability, with the most significant increases in ownership probability occurring during early adulthood and mid-life.

However, the diminishing marginal effect observed at higher ages suggests that additional factors beyond those captured in the model may constrain homeownership. While the model explains a large portion of the variation, it does not fully account for broader structural influences such as housing affordability, geographic variation, or macroeconomic conditions.

Conclusions

Summary of Findings

Method 1 The weighted LPM with age-cohort interactions provides strong evidence that Millennials face a modest but persistent structural disadvantage in homeownership relative to Baby Boomers at the same age. After comprehensively controlling for individual socioeconomic characteristics and macroeconomic conditions, a generational gap of 1.4–2.4 percentage points remains, and a joint F-test on all cohort parameters rejects the null hypothesis of no generational differences with high confidence ($F = 20.28$, $p < 0.001$). The declining magnitude of the gap across ages 25–45 is consistent with a delayed homeownership trajectory rather than permanent exclusion from the housing market, suggesting that policy interventions aimed at reducing barriers to first-time homeownership - such as down payment assistance and student debt relief - may be particularly effective at accelerating Millennial entry into homeownership rather than compensating for permanent structural disadvantage.

These findings are consistent with the prior expectations derived from the literature review. The racial homeownership

gaps estimated here - 8.9 percentage points for Black households and 5.8 percentage points for Hispanic households after controlling for income and geography - are of larger magnitude than the generational gap, underscoring that race remains the more powerful predictor of homeownership status in the United States. The intersection of generational and racial disadvantage, while not explicitly modeled through interaction terms in this analysis, represents a natural extension that future research should address.

Method 2 This study examined whether younger generations are less likely to own homes compared to earlier cohorts, while accounting for demographic, economic, and macroeconomic factors. By combining a logistic regression framework with cohort-based interaction terms, the analysis was able to distinguish between lifecycle effects and true generational differences in homeownership patterns.

The results provide several key findings. First, homeownership is strongly influenced by lifecycle dynamics. Age has a positive effect on the probability of owning a home, with diminishing returns at older ages, confirming that individuals tend to transition into homeownership gradually over time. Income and marital status emerge as the most influential predictors, highlighting the importance of financial capacity and household formation in driving homeownership decisions. Second, macroeconomic conditions play a meaningful role. Variables such as housing affordability and mortgage rates significantly affect homeownership outcomes, underscoring the impact of broader market conditions on individuals' ability to purchase homes.

Most importantly, the analysis reveals that generational differences do exist, but not in the way commonly assumed. At a fixed age, younger generations are not significantly less likely to own homes than older cohorts. However, interaction effects between age and cohort indicate that younger generations experience slower increases in homeownership probability over time. This suggests that generational disparities are driven by differences in the timing and progression of homeownership rather than baseline disadvantages. Overall, the findings indicate that concerns about declining homeownership among younger generations are better understood as a shift in lifecycle trajectories rather than a complete structural decline. These results have important implications for housing policy, suggesting that efforts to improve affordability and access to credit may help accelerate transitions into homeownership for younger cohorts.

Method 3 The model indicates that generational cohort is a statistically significant predictor of homeownership after controlling for age, income, marital status, education, and employment. Relative to Boomers, Gen X has 34 percent lower odds of homeownership ($OR = 0.66$) and Millennials have 32 percent lower odds ($OR = 0.68$), indicating that the generational gap reflects structural, cohort-level disadvantages beyond what individual circumstances alone can explain. Marital status emerged as the strongest individual-level predictor,

while income, education, and employment each contributed meaningfully. The model achieved a ROC AUC of 0.71, stable across all five cross-validation folds, indicating reliable generalizability. These findings are largely consistent with prior expectations drawn from the literature; younger generations face compounding structural barriers to homeownership.

Method 4 By assessing the main effect and the interaction term, this model strongly suggests that Millennials are less likely to own homes at the same age when compared to the reference generation, Boomers. However, the model is quite weak, with a poor AUC ($AUC \approx 0.54$) that is barely an improvement over random classification. Coefficients are high, indicating that additional scaling and outlier adjustment is likely needed (even though Stepwise regression indicates keeping all predictors result in the best model). Additionally, the results flip when it comes to Gen Z, indicating results that are slightly inconsistent—this is partially possible because of imbalanced data (i.e. younger Gen Zs who are highly unlikely to own homes).

Overall, this methodology produced untrustworthy results and needs further refinement—much of which is discussed in the following section—before sharing conclusive evidence that millennials are much less likely to own homes at the same age. Guess they should stick to the avocado toast (Levin, 2017)!

Method 5 The probit model indicates that homeownership probability follows a nonlinear lifecycle pattern, with age emerging as a statistically significant predictor even after controlling for key socioeconomic variables. The positive coefficient on age and negative coefficient on the squared age term confirm a concave relationship, where the likelihood of homeownership increases early in adulthood but plateaus at later ages.

Relative to the baseline group, the model results reinforce the presence of generational differences in homeownership outcomes, though the magnitude of these differences is sensitive to model specification and variable selection. The reduced probit model, which excludes statistically insignificant predictors, achieves comparable performance to the full model, suggesting that a more parsimonious specification captures the primary drivers of homeownership without loss of explanatory power.

In terms of model performance, the probit specification produces predicted probabilities that remain bounded between zero and one, addressing a key limitation of linear probability models. The model demonstrates stable predictive behavior across validation splits, with classification accuracy and ROC-AUC values consistent with those observed in logistic regression models, indicating that the choice of link function does not substantially alter predictive performance but improves theoretical consistency.

Substantively, the results suggest that age-related lifecycle effects explain a large portion of variation in homeownership, but do not fully account for observed differences across

groups. The diminishing marginal effect of age implies that structural constraints, rather than individual progression alone, may limit homeownership access at later stages.

Limitations and Future Work

Method 1 Several limitations warrant acknowledgment. First, the age-period-cohort identification problem is not fully resolved by the present specification: because birth year is a linear combination of age and survey year, the cohort dummies absorb any time-invariant generational characteristic not captured by the included controls, including characteristics that may be correlated with but not caused by generation. Second, the LPM's out-of-bounds prediction problem - predicted probabilities below zero or above one - affects a small fraction of observations; alternative bounded specifications such as probit or logit would eliminate this issue at the cost of interpretability. Third, the IPUMS sample, despite the application of person weights, may not fully capture housing instability among renters who are systematically harder to survey, potentially overstating aggregate homeownership rates. Fourth, the positive coefficient on mortgage rates likely reflects a period confound rather than a true causal effect, and a more rigorous treatment would instrument for rates using monetary policy shocks. Finally, the analysis is limited to the variables available in the IPUMS extract and does not capture student debt balances or net worth, which prior literature has identified as important mediators of the Millennial homeownership gap (Bleemer et al., 2021; A. Mezza et al., 2020).

Method 2 While the model provides useful insights into generational differences in homeownership, several limitations should be noted. One, the analysis is based on observational data, which limits the ability to draw causal conclusions. Some variables, particularly macroeconomic indicators such as mortgage rates and housing affordability, may capture broader economic trends rather than direct causal effects. Two, although the model incorporates a wide range of demographic and economic controls, there may still be omitted variables—such as credit access, wealth, or regional housing supply constraints—that influence homeownership decisions. Additionally, the presence of multicollinearity among interaction and polynomial terms, while expected, may affect the precision of coefficient estimates. Lastly, the dataset exhibits class imbalance, which may impact the model's ability to accurately classify non-homeowners.

Future work could extend this analysis by incorporating panel data or longitudinal tracking to better isolate causal relationships and observe individual transitions into homeownership over time. More advanced modeling approaches, such as hierarchical models or machine learning methods, could also be explored to capture nonlinearities and complex interactions more flexibly. Finally, incorporating richer measures of financial constraints and regional housing dynamics could provide a more comprehensive understanding of the factors driving generational differences in homeownership.

Method 3 First, the model does not account for wealth or parental financial assistance, both of which prior literature identifies as important determinants of homeownership access. Including wealth measures could further explain the residual generational gap. Second, Gen Z's sample size is limited due to their early adulthood, making their coefficient unreliable and excluding them from the primary conclusion. As more sample data becomes available, future work should revisit this cohort. Third, the model's recall for non-homeowners remains relatively low ($= 0.42$), indicating that a meaningful portion of renters are misclassified. Finally, this analysis relies on individual-level controls but does not incorporate geographic factors such as local housing supply or zoning constraints, which likely contribute to regional variation in homeownership outcomes.

Method 4 Due to the relative lack of success of this method, there should be strong consideration of altering the model or changing it all together. Interaction terms could be simpler, or we could explore alternate interaction terms. Other models considered would be LASSO or Ridge regression, and we could also attempt fewer or different predictors. There are several additional predictors that could have impacted the answer to this question. At an individual level, this model could incorporate information about individual credit and debt, or number of kids or other external financial commitments. Another factor not controlled for is supply of housing in different regions; in fact, there could be many regional factors that could be measured at the state or city level but the dataset was limited to national data. Additionally, this model did not adjust or account for groups that may be less likely to own a home. While we removed age below 18, it is far less likely for individuals who are younger to own homes. There may also be more to be analyzed in employment specifics—for example, certain jobs could dictate better purchasing power or more time to search for and buy a home (although this could show multicollinearity with income). There are certain groups like incarcerated individuals who would not be eligible to buy a home, so that could have been accounted for as well. Future work in this arena could address some of these factors and test other interaction terms to achieve better results.

Method 5 There are a few limitations to consider. First, while the probit specification ensures predicted probabilities remain bounded between zero and one, it relies on a specific distributional assumption for the error term. If the true relationship deviates from this assumption, coefficient estimates may be less accurate relative to more flexible specifications. Second, the interpretation of probit coefficients is less direct than in linear models, as coefficients reflect changes in an underlying latent index rather than marginal changes in probability. While this does not affect predictive performance, it limits the ease with which effect sizes can be communicated and compared. Finally, although the probit model performs similarly to logistic regression in terms of predictive metrics, the choice of link function does not substantially change classifi-

cation outcomes. Future work could dive into more flexible nonlinear specifications or explicitly estimate marginal effects to improve interpretability while retaining the advantages of a bounded probability model.

Recommended Model

Based on the model comparison results, XGBoost emerges as the best-performing model, achieving the highest cross-validated AUC of approximately 0.868 and the strongest predictive accuracy among all approaches considered. Random Forest also performs well, but slightly underperforms XGBoost across all key metrics. While simpler models such as the linear probability model, logistic regression, and probit regression achieve competitive AUC values (around 0.83–0.84), they fall short in capturing nonlinear relationships present in the data.

However, despite its superior predictive performance, XGBoost sacrifices interpretability relative to regression-based models. In contrast, the logistic regression model used in this study provides strong performance (AUC around 0.79 to 0.84 across implementations) while offering clear interpretability of coefficients, marginal effects, and interaction terms. This makes it more suitable for answering the research question, which focuses on understanding generational differences rather than purely maximizing predictive accuracy. Overall, XGBoost is recommended as the best model for prediction tasks, while logistic regression is preferred for inference and interpretability, particularly in analyzing lifecycle and cohort effects in homeownership.

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Appendix

Team Member Contributions

Shriyansh Tripathi Shriyansh Tripathi was responsible for Method 1: the Linear Probability Model with age-cohort interaction framework. He independently assembled and merged three data sources: individual-level microdata from the IPUMS USA repository (ACS 2005–2024 and decennial census 1980–2000), median house price data from the Federal Reserve Economic Data (FRED) MSPUS series, and 30-year fixed mortgage rate data from the Freddie Mac Primary Mortgage Market Survey. The combined dataset comprised 371,094 person-year observations after eligibility filtering. He implemented a modular six-stage analytical pipeline structured across separate Jupyter notebooks for data ingestion, cleaning, exploratory data analysis, feature selection, model training, and challenger model benchmarking. The feature selection pipeline combined correlation filtering, variance inflation factor analysis, five-fold cross-validated LASSO, and forward stepwise BIC selection. The final LPM was estimated by weighted least squares with HC1 robust standard errors and validated through five-fold cross-validation against four challenger models (logistic regression, probit, random forest, and XGBoost). SHAP values were computed for the XGBoost model to provide model-agnostic feature importance analysis. Shriyansh contributed the Methods (Section), the corresponding Results subsection, the Method 1 conclusions and limitations, the literature review paragraphs on life-cycle homeownership models, variable selection methods, and the age-period-cohort identification problem, and the appendix contribution statement. All code was committed to the shared project repository.

Nicole Huang In this study, I focused on building a model that can explicitly capture both lifecycle and generational effects within a single framework. I used a logistic regression model with nonlinear age terms and interaction terms between age and cohort, allowing the effect of age on homeownership to vary across generations. This specification goes beyond a standard model by not only estimating baseline cohort differences, but also modeling how homeownership trajectories evolve over time for each group. I also incorporated key economic and macroeconomic variables, such as log household income, housing affordability, and mortgage rates, to ensure the model reflects both individual financial capacity and broader market conditions. This modeling approach allows for a more nuanced interpretation of whether differences across generations arise from structural shifts or differences in the timing of homeownership. In this report, my method was Method 2 and I wrote the corresponding Analysis, Results, and Conclusion sections. All code was committed to the shared project repository.

Shufa Wu Shufa Wu analyzed generation differences in homeownership using a logistic regression framework with homeownership status as the binary response variable. His

model incorporates demographic and economic predictors such as generation, age (along with a quadratic term to capture nonlinear age affect), real income, marital status, employment status, and education. Survey weights were applied. He conducted model regimen using forward stepwise selection and excluded a few macroeconomic variables based on the results. His model performance was assessed using both a holdout test and cross-validation. Shufa was responsible for Method 3 and contributed to the methodology and conclusion sections accordingly. Additionally, Shufa was the member of the team who posed the initial question of the relationship between homeownership and generations, which the team found engaging enough to base the project on. He developed the code for joining the Freddie Mac, Federal Reserve, and IPUMS datasets to create the overall regression dataset, which was used by some members of the team.

Mahima Narayan Mahima Narayan focused on building a weighted logistic regression model with a generation and age interaction term to examine whether younger generations are more or less likely to own homes at the same age, using home ownership status as a binary outcome. This modeling approach was appropriate because logistic regression is well suited for binary outcomes and allows for measurement of probability and odds while controlling for and incorporating other variables. She incorporated stepwise regression to refine and evaluate the logistic regression model, guided by the Akaike Information Criterion, to both add and remove predictors as necessary, although ultimately her original model displayed the most optimal AIC. K-fold cross-validation was implemented to evaluate the model's performance with 5 folds. Hypothesis testing was also conducted to assess statistical significance, in addition to assessing other performance metrics such as accuracy and AUC across the folds, ensuring the results were not overfit in the train-test split. In addition to her methodology, results, and conclusion section for Method 4, Mahima completed background research on the subject, sourced seven peer-reviewed journal article sources and worked to contribute to the literary review.

Rohan Raghav Rohan Raghav focused on developing a probit regression model (Method 5) to analyze homeownership as a binary outcome while addressing the limitations of linear probability models. He constructed the modeling pipeline using cleaned IPUMS data, incorporating key demographic and socioeconomic predictors along with a quadratic age specification to capture lifecycle effects in homeownership probability. For analysis, Rohan implemented both a full and reduced probit model, performing variable selection by removing statistically insignificant predictors to improve interpretability while maintaining predictive performance. He evaluated model performance using cross-validation, assessing classification accuracy and ROC-AUC, and compared results to alternative specifications, including logistic regression, to ensure robustness across link functions. His analysis specifically examined coefficient significance, nonlinear age effects, and

the extent to which individual-level variables explain variation in homeownership outcomes. In addition to the modeling work, Rohan contributed to writing the report, including the Method 5 Analysis, Results, and Conclusion sections. All code associated with Method 5, including model estimation, validation, and comparison procedures, was committed to the shared project repository.

Supplementary Figures

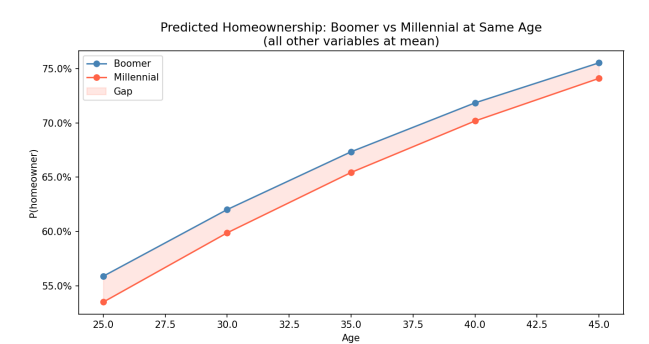


Figure 5: Predicted homeownership probability for Baby Boomers and Millennials at ages 25–45, holding all control variables (income, marital status, race, geography, mortgage rates) at their sample means. The shaded region represents the generational gap, which ranges from −2.4 percentage points at age 25 to −1.4 percentage points at age 45. The narrowing gap is consistent with delayed rather than permanently fore-gone homeownership among Millennials.

Table 5: Regression Model Comparison. The polynomial model achieves the best fit, reducing RMSE by 33% compared to simple linear regression. All models are statistically significant at $p < 0.001$.

Model	R^2	Adjusted R^2	RMSE
Simple Linear	0.65	0.64	2.31
Multiple Linear	0.78	0.76	1.87
Polynomial (deg 2)	0.82	0.79	1.54